

Maximo Jimenez
ENGR 214 - Statics - Vereen
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Individual Contribution Report

I can start this by simply listing (most) of my contributions to all *written* work for the bridge project starting with the bridge design memo.

Design Memo:

- Proposed the triangle truss and provided the first mockup
- Proofread, edited, and formatted the document

Design Memo #2:

- Contributed several passages to the “V2 iteration” section of the report.
- Contributed several passages to the “Statics Analysis and Reasoning” section
- Wrote the conclusion
- Formatted the document, added diagram labels, edited/proofread all sections.

Truss Test Report

- Wrote the “Load at Failure and Failure Mode” section
- Wrote the “Decision Tree Matrix” section
- Performed a general proofread/editing pass on all sections

Bridge Proposal

- Wrote the “Lessons Learned from Truss Designs” section
- Proofread and lightly edited all sections

Build Log + Updated Budget Summary

- Not gonna lie I think Timothy did most of this. He told us groupmates it was no biggie and it wasn't a ton of work.

Failure Analysis Report

- Wrote the “Assumptions” section
- Wrote the “Discussion—Results” section
- Wrote the “Discussion — Failure Analysis” section
- Proofread and general edit

Final Technical Report

- This one is difficult to quantify. I did a LOT of the initial formatting, copying/pasting etc. and then continued to try and reduce redundancy in the content by pruning the report. I then went through and introduced new passages in many places to narratively bridge the report. A large part of this was flow-ey/stylistic in nature and non-technical. I also went through and ironed the doc to maintain consistent past/present tense.

Beyond the written work it is again, difficult to quantify exactly what I did every lab day for the bridge project. I think I was generally engaged, proactive, and productive in lab days. I contributed ideas and labor in the form of fabrication, performing tests, etc.

As for my role in the team, I think we decided on me being the ‘materials’ guy at the beginning. At the beginning of the project, I sourced all of the metal nuts and bolts from Monroe Hall for the preliminary design. Later, I ended up sourcing the lightweight fasteners we used in our final design, so hopefully I did a decent job. Beyond that, I think I was an active contributor in the design and construction process throughout the project.

Reflection

In this project I learned what it’s like to work in a high-functioning, outcome oriented engineering group. It was really awesome to see how each team member contributed in their own way to the overall success of the team.

One thing I wish I knew at the start of the project would be the overall scoring criteria for the bridge. I understand that the smoke and mirrors regarding that was partially by design, but it definitely would have helped to have a concrete understanding of the *entirety* of the challenge criteria, not just the ‘critical criteria’, so-to-speak.

If I were to do this project again, I would’ve changed the loading mechanism we used on the final test day. With a better loading system, I wonder if the bridge would’ve broken at all. I applied approximately 40lbs of force to just one truss after truss test day and it was fine. Design-wise, I actually quite like the direction we took. If I had to change something else, maybe it would be lengthening the ‘feet’ of the bridge in order to ensure flush contact with the table.

In regards to working with my team, I don’t think we had any zero members (zing!). Everyone played important roles and satisfied their responsibilities without fail. I think the only thing we could have done better would be time management maybe? The thing is, a lot of the teammates were working jobs in addition to school. It’s hard to get things to line up *every time*. I think, given the circumstances, we did a more than fair job. Honestly, my only “regret” is not getting to work with them on more projects!